TEKOLO YA DILOKI TŠA DITŠHELETE

Test with Real-world Budget Inputs Across Domains

Puku ya Tekolo le Phatlhalatšo ya Ditekanyetso tša Ditšhelete tša Nnete go Dikarolo tša Phapano

Mofuta wa Tokomane: Molaotheo wa Tekolo ya Diloki tša Software le AI

Moroeroe: Real-world Budget Allocation & Stress Testing

Abori: Injenere tša Software, Bahlahlobi ba Ditšhelete le Baakanyi ba Diloki

Seemo: E Amogetswe ka Semolao / Approved Standard

1. Matseno go Tekolo ya Budget ya Nnete

Tekolo ya diloki tša ditšhelete (budgeting systems) e hlokagala kudu go netefatša gore ditshepediše tša khowebo le AI di kgona go šoma ka nepo ge di lebeletšane le dipalopalo tša nnete. Go šomiša dintlha tša nnete (Real-world Budget Inputs) go thuša go hwetša diphošo pele tshepedišo e ka ya go badiriši.

1.1 Bohlokwa ba go Leka ka Dintlha tša Nnete

Ge re leka diloki tša ditšhelete go dikarolo tša go fapana (Domains), re swanetše go lebelela dintlha tša go fapafapana tša ditšhelete:

Dintlha tša Bohlokwa tša Tekolo:

- **Phapano ya Ditšhelete (Currency Multi-variability):** Go leka ditshepedišo ka ditšhelete tša go fapana (USD, EUR, ZAR, INR).
- **Mekgwa ya Ditjeho (Cost Structures):** Ditjeho tša go se fetoge (Fixed) le tša go fetoga (Variable costs).
- **Diphetogo tša Nako (Dynamic Constraints):** Diphetogo tša dipeelo tša ditšhelete go latela nako le dikgwedi.

2. Tekolo go Lephata la Maphelo (Healthcare Domain)

Lephateng la maphelo, ditekanyetso tša ditšhelete di na le melao e tiilego le ditjeho tša go fapafapana tša dikalafo, didirišwa le baoki.

2.1 Mefuta ya Dintlha tša Healthcare Budget

```
{
  "domain": "Healthcare",
  "facility_id": "HOSP-2026-001",
  "annual_budget_usd": 12500000.00,
  "categories": {
    "medical_supplies": 4500000.00,
    "staff_salaries": 5500000.00,
    "emergency_fund": 1500000.00,
    "equipment_maintenance": 1000000.00
  },
  "constraints": {
    "min_emergency_percent": 10.0,
    "regulatory_audit_compliant": true
  }
}
```

3. Tekolo go Lephata la Thekišo (E-Commerce & Retail)

E-commerce e hlokagala tekolo ya ditekanyetso tša ditšhelete go latela dinako tša thekišo e ntši (peak seasons) le ditjeho tša bapapatši.

3.1 Model ya Dintlha tša E-Commerce

```
{
  "domain": "E-Commerce",
  "campaign_name": "Black Friday Sale 2026",
  "total_budget": 500000.00,
  "channels": [
    {"name": "Google Ads", "allocated": 200000.00, "max_cpc": 2.50},
    {"name": "Social Media", "allocated": 180000.00, "max_cpc": 1.80},
    {"name": "Influencer Marketing", "allocated": 120000.00, "fixed_cost": true}
  ]
}
```

4. Tekolo go Lephata la Theknolotši ya Leru (Cloud IT)

Go laola ditekanyetso tša ditšhelete tša Cloud (AWS, Azure, Google Cloud) go hlokagala tekolo ya nnete go thibela go tsena moseleng ka ditjeho tša go sa letelwa.

4.1 Cloud Budget Input Test Specification

```
{
  "domain": "Cloud Infrastructure",
  "monthly_cap_usd": 75000.00,
  "services": {
    "compute_ec2": {"budget": 35000.00, "auto_scale": true},
    "database_rds": {"budget": 20000.00, "multi_az": true},
    "storage_s3": {"budget": 10000.00, "data_retention_days": 365},
    "networking": {"budget": 10000.00}
  },
  "alert_thresholds": [50, 75, 90, 100]
}
```

5. Tekolo go Lephata la Thuto le Dinyakišišo

Ditekanyetso tša thuto di tshetletswe go dithuso tša ditšhelete (grants) tšeo di nang le melao e kgethegileng ya go dinomoro.

Melao ya Dithuso tša Thuto:

- **Grant Allocation Rules:** Chelete e swanetše go šomišwa fela go ditshenyagelo tše di dumeletsweng.
- **Audit Requirements:** Pegelo ya ditšhelete e swanetše go tswa kgwedi le kgwedi.

6. Go Hlama Dintlha tša Tekolo (Synthetic Data Generation)

Ge go se na dintlha tša nnete ka baka la go babalela sephiri (privacy laws), re hlama dintlha tša go swana le tša nnete go leka ditshepedišo.

6.1 Python Script ya go Hlama Budget Inputs

```
import random

def generate_realworld_budget(domain, base_amount):
    variation = random.uniform(0.85, 1.25)
    final_budget = round(base_amount * variation, 2)
    return {
        "domain": domain,
        "input_budget": final_budget,
        "tax_margin": round(final_budget * 0.15, 2),
        "operational_reserve": round(final_budget * 0.05, 2)
    }

print(generate_realworld_budget("FinTech", 1000000))
```

7. Stress Testing le Maemo a Magolo (Edge Cases)

Go leka tshepedišo ka maemo a magolo go go thuša go bona ge e ka phetsega goba ya palelwa.

Lekhetlo la Tekolo	Dintlha tša Inputs	Poelo e Letetšwego
Zero Budget	`budget = 0.00`	Validation Error (422 Bad Request)
Extreme Overflow	`budget = 999,999,999,999.99`	Handling Big Numbers ka nepo
Negative Value	`budget = -5000.00`	Reject Input ka molaetsa wa phoso

8. Pipolene ya Tekolo ya Boisao (Automated Testing Pipeline)

Go kenya ditekolo tša ditekanyetso tša ditšhelete go CI/CD pipeline go netefatša gore khoutu e mpsha ga e senye ditshepedišo tša kgale.

```
# Pytest Integration Test
def test_realworld_budget_validation():
    payload = {"domain": "Logistics", "budget": 250000.00}
    response = client.post("/api/v1/budget/validate", json=payload)
    assert response.status_code == 200
    assert response.json()["is_valid"] == True
```

9. Kakaretso le Ditsela tše Kaone (Best Practices)

Tekolo ya ditekanyetso tša ditšhelete ka dintlha tša nnete go dikarolo tša go fapana e hlokagala go netefatša gore software e šoma ka botlalo le go thibela tahlego ya ditšhelete.

Ditsela tše Kaone (Best Practices Checklist):

- Šomiša dintlha tša nnete tša di-domain tša go fapafapana (Healthcare, Retail, Cloud, Tech).
- Leka maemo a magolo (Edge cases: Zero, Negative, Massive Numbers).
- Dira gore ditekolo di sepedišwe ka boona ho CI/CD pipeline.